NaviCore

Maritime Operations Command Platform

CodeBuddy × GLM Global AI Hackathon Singapore 2026

NaviCore is a maritime operations command platform for vessel fleet management. It unifies fleet visibility, cargo planning, weather awareness, crew operations, fuel and cost optimization, and intelligence monitoring in a single control surface.

Project Name	NaviCore
Team	SYED AZHAR BIN SYED JAKIRUL ALAM, THANAKESH SRI RAU A/L SUBRAMANIAM, LIA
Deployment URL	https://06aa3b79afa942baa5833880a4798af1.ap-singapore.myide.io

Problem & Solution

Maritime operations are often fragmented across disconnected tools: route visibility in one place, cargo balancing in another, and threat intelligence in separate dashboards.

NaviCore consolidates mission-critical workflows into one platform. Operators can monitor vessel movement, optimize cargo loading with an AI-assisted balancing algorithm, assess weather risk, manage crew readiness, and monitor operational costs in real time while tracking global maritime-relevant intelligence.

Core Value

• Single pane of glass for maritime operations

• AI-assisted cargo arrangement with operational safety goals

• Faster response to weather and geopolitical risk signals

• Improved voyage efficiency through fuel and cost insights

Product Features

Fleet Map: Live vessel tracking with AIS layer, port overlays & trade routes.

Cargo Arrangement: Interactive deck slot grid with AI-powered load balancing (Center-Out, Bottom-Up algorithm).

Weather Risk: Real-time weather radar and voyage risk assessment.

Crew Roster: Crew management per voyage.

Fuel Optimizer: Voyage fuel cost analysis.

Cost Ledger: Real-time cost accrual engine.

Live Intelligence: Aggregated global news feed with threat classification (military, cyber, nuclear).

Deckhand App: Mobile QR scanner for cargo loading status updates.

Architecture & Workflow

NaviCore uses a React-based command UI and backend services for cargo, voyage, and intelligence processing. The cargo optimizer prioritizes bottom-up vertical stability, center-out bay distribution, and port-starboard pairing.

Operational flow: select voyage → review manifest → run AI optimize → validate slot proposals → confirm/rearrange placements → save final layout.

Demo & Submission Summary

Live deployment: https://06aa3b79afa942baa5833880a4798af1.ap-singapore.myide.io

Generated by CodeBuddy for Hackathon submission.